

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I swami.sujit(192525031)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: s sujit_____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Name: S. SWIL

Reg. No. : 192525031

Course code : CSA1507 COI-ATT1

Class Test

Q1) Cloud computing:

Cloud computing is a computing services which help us to store the extra data ~~bit~~ within the same computer without using external devices. And it gives services like (software, web browsers, websites etc).

i) on-demand self services:

It is a service in which we can get the extra storage by ourselves without contacting provider.

ii) Broad network access:

With this service we can access a broad networks like (software, websites etc).

iii) Resource Pooling: It is the service which helps us in data storing and store it securely.

iv) Rapid elasticity:

This service is rapidly evolved and helps us using the rapid retrieving data all over the services and storing the data rapidly.

Q2) Evolution :

- i) Common Computer ^(1960's) : There are only one computer And it was used by many users.
- ii) Personal Computers ^(1980's) : It's the time where all have their personal servers.
- iii) Client Based Servers (1995) : client's are requested for a servers for the employment.
- iv) Virtualization ^(2000's) : everyone got the role of using internet & web sites And web browsers.
- v) Cloud Computing (2010's) : everyone started using computing services for storing their extended data.
- * Conclusion : This was the evolution done from mainframe to cloud computing.

Q3) Virtualization extended in cloud computing. It plays a major role in cloud computing It helps to improve the performance.

Virtualization

- It helps to run large servers.
- Runs multiple OS servers
- foundation technology

cloud computing

- It is used for computing services.
- Provides storage, servers
- complete serving model.

* Conclusion : At the last virtualization extend in cloud computing.

94)

* Data centers: Data centers are used to store users data safely. And deliver the data to user when it is useful. And when the data centers are heated cooling will be done in the data centers with coolants.

* Role of Data centers in cloud computing:
→ Data centers help users to store data for cloud computing. Without them the process will be ruined because the main theme of cloud computing is to store the extended data & use it effectively so without any data cloud computing can't run.

* Conclusion: Data centers plays a main role in cloud computing.

015)

* IaaS (Infrastructure as a service):

→ This service depends on the infrastructure such as machines, web browsers E.T.C.

* Ex: AWS EC2.

* PaaS (Platform as a service):

→ This service depends on platforms such as software, websites, apps E.T.C.

* Ex: Google one.

* SaaS (Software as a service):

→ This service gives everything to use software, and provider manages everything and user can use the software.

* Ex: Gmail.

* XaaS (Everything as a service):

→ We can use everything as a service in this service. we can use software

(or) Platform at once.